Python Stock Analytics Project: Strategy Design and Backtesting

Student Name: Crystal Garcia

Date: May 13, 2026 (revised October 12, 2026)

Instructor: Yongzhi Wang, PhD

1. Topic Selected

I chose to work on topic 1 for this analysis because it provides a clear framework for evaluating whether an active, rule-based technical strategy can effectively manage risk and improve performance compared to a passive "Buy-and-Hold" benchmark.

2. System and Data Assumptions

The following parameters were set for the backtesting environment and dataset.

- **Data Source:** Historical market data was retrieved from Yahoo Finance via the yfinance library.
- **Selected Stock Universe:** Five U.S. stocks were selected for analysis: AAPL, MSFT, AMZN, GOOGL, and NVDA.
- **Time Period:** The backtest covers a four-year duration from **January 1, 2020, to January 1, 2024**. Prices from 2019 were also downloaded so the 200-day moving average is available on the first trading day of 2020; performance is measured only from January 2, 2020.
- **Data Frequency:** Daily adjusted closing prices were utilized to account for dividends and stock splits.
- **Starting Capital:** An initial balance of $10,000 was allocated to each stock test.
- **Transaction Logic:** The system assumes a "Long Only" approach with no transaction costs or commissions, and no interest earned on cash.
- **Negative Balance Constraint:** As required, the portfolio balance can never drop below zero. The strategy is long-only with no leverage or short selling, so each day the portfolio is either fully invested in the stock or fully in cash, and its value can fall at most to zero.

Table 1: Sample of the raw AAPL dataset downloaded from Yahoo Finance, showing daily price and volume data before strategy calculations.

Price	Open	High	Low	Close	Volume
Date					
2020-01-02	71.344062	72.394093	71.091191	72.333885	135480400
2020-01-03	71.563228	72.389280	71.406689	71.630661	146322800
2020-01-06	70.754006	72.239935	70.503539	72.201401	118387200
2020-01-07	72.211033	72.466315	71.642674	71.861832	108872000
2020-01-08	71.565606	73.318862	71.565606	73.017822	132079200

This dataset was used as the input for calculating moving averages, generating signals, and performing backtesting analysis.

3. Strategy Description

Moving Average Crossover Strategy was implemented in this analysis.

- **Buy Condition:** Enter a position when the **50-day Simple Moving Average (SMA)** crosses above the **200-day SMA** (a "Golden Cross").
- **Sell Condition:** Exit the position and move to cash when the 50-day SMA crosses below the 200-day SMA (a "Death Cross").
- **Objective:** This strategy attempts to capture major upward trends while avoiding severe market downturns by exiting to a cash position.



4. Backtesting Methods

The backtest was developed in Python using pandas, matplotlib, and yfinance.

1. **Preprocessing:** Data was cleaned by removing missing values. The 2019 prices serve only as a warm-up period for the moving averages; the data is then trimmed so both portfolios start with $10,000 on January 2, 2020 and are compared over the same dates.
2. **Signal Generation:** Rolling means were calculated, and signals were shifted by one day to avoid "look-ahead bias".
3. **Simulation:** Portfolio value was calculated daily by applying strategy returns to the previous day's capital.
4. **Benchmark:** A "Buy-and-Hold" return was calculated as a primary baseline for comparison.

5. Backtesting Result Analysis

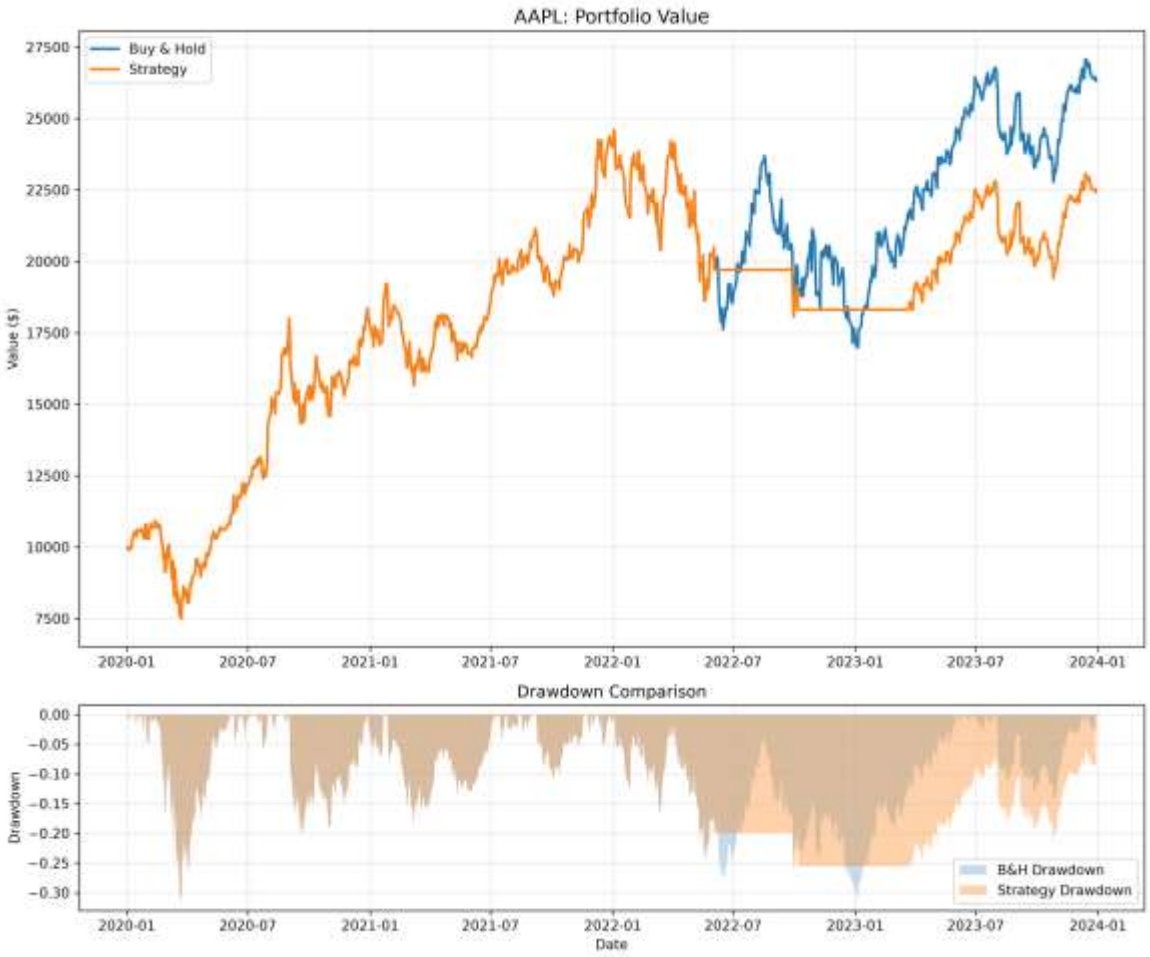
I evaluated performance using total return, annualized return, maximum drawdown, and positive-day percentage (the share of invested days that ended with a gain).

Numerical Summary Table

	B&H Total %	Strategy Total %	B&H Ann. %	Strategy Ann. %	B&H Max DD %	Strategy Max DD %	B&H Positive Day %	Strategy Positive Day %
AAPL	163.19	124.23	27.43	22.42	-31.43	-31.43	52.69	52.83
MSFT	142.95	160.18	24.90	27.07	-37.15	-28.04	52.74	54.53
AMZN	60.10	68.66	12.51	13.99	-56.15	-25.31	51.94	52.67
GOOGL	104.12	125.02	19.57	22.53	-44.32	-30.87	53.49	56.01
NVDA	728.90	823.59	69.86	74.53	-66.34	-37.55	54.44	55.30

Visual Comparisons

AAPL_results image



AAPL is the one stock where Buy-and-Hold finished ahead (+163% vs. +124%). The strategy sold in June 2022, briefly re-entered in the fall, and did not re-enter for good until March 2023, missing part of AAPL's rebound. Its maximum drawdown equaled Buy-and-Hold's (31%) because both portfolios were invested during the March 2020 crash.

GOOGL_results image



For GOOGL, the strategy moved to cash in March 2022 and re-entered in May 2023, finishing ahead of Buy-and-Hold (+125% vs. +104%) with a smaller maximum drawdown (31% vs. 44%).

NVDA_results image



For NVDA, the strategy sat in cash from April 2022 to January 2023, avoiding most of the 66% decline. It finished ahead of Buy-and-Hold (+824% vs. +729%) with a much smaller maximum drawdown (38% vs. 66%).

Key Observations

- **Risk Mitigation:** The strategy's maximum drawdown was lower than Buy-and-Hold's for MSFT, AMZN, GOOGL, and NVDA, and equal for AAPL (31%). It avoided much of the long 2022 decline, but not the sudden March 2020 crash: the 50/200-day averages react too slowly, and that crash was the strategy's largest drawdown for AAPL, MSFT, GOOGL, and NVDA.
- **Performance vs. Benchmark:** The strategy produced higher total returns than Buy-and-Hold for 4 out of 5 stocks (MSFT, AMZN, GOOGL, and NVDA). AAPL was the exception.
- **Stability:** The visualizations show that the strategy creates a "smoother" equity curve by staying in cash during sustained market corrections.

6. Conclusion

From 2020 to 2023, the Moving Average Crossover strategy improved both total return and maximum drawdown for four of the five stocks, mainly by moving to cash during the long 2022 decline in technology stocks. Its weaknesses are speed and false signals: it could not protect against the sudden March 2020 crash, and for AAPL a brief false re-entry and a late return to the market left it behind Buy-and-Hold. These results come from a single four-year period that included one long, gradual bear market, which favors trend-following rules, and from five large technology stocks that performed well. The strategy is best suited for investors who want to limit losses in prolonged downturns, with the understanding that it will not prevent losses in sudden crashes.

Revision note: The original version of this report downloaded data starting on January 1, 2020, so the 200-day moving average was not available until October 2020 and the strategy stayed in cash until then while Buy-and-Hold was invested. This made the strategy appear safer (it skipped the March 2020 crash) and less profitable than it really was. This revision adds 2019 data as a warm-up period so both portfolios are compared fairly from the same start date, and all results, charts, and conclusions have been updated.

7. AI Claim

I utilized AI tools ChatGPT and Gemini, to assist with this project. These tools were used to help structure the backtesting Python logic, debug data-alignment issues, and organize the formatting of the final report. All code logic and analytical conclusions were reviewed and verified before final submission. For this revision, Claude (Anthropic) was used to review the analysis, identify the moving-average warm-up issue, and help update the code and report.